Ortho and para hydrogen are similar in chemical properties but differ in some of the physical properties. For example, the melting point of para hydrogen is 13.83 K while that of ortho hydrogen 13.95 K; boiling point of para hydrogen is 20.26 K while that of ortho hydrogen 20.39 K. Since the nuclear spins are in opposite directions the magnetic moment of para hydrogen is zero and ortho hydrogen has magnetic moment twice that of a proton.

## 4.2 Preparation of Hydrogen

High purity hydrogen (>99.9 %) is obtained by the electrolysis of water containing traces of acid or alkali or the electrolysis of aqueous solution of sodium hydroxide or potassium hydroxide using a nickel anode and iron cathode. However, this process is not economical for large-scale production.

At anode : $2\ OH^- \rightarrow H_2O + \frac{1}{2}\ O_2 + 2e^-$

At cathode : $2\ H_2O + 2\ e^- \rightarrow 2\ OH^- + H_2$

Overall reaction : $H_2O \rightarrow H_2 + \frac{1}{2}\ O_2$

### 4.2.1 Laboratory Preparation

Hydrogen is conveniently prepared in laboratory by the reaction of metals, such as zinc, iron, tin with dilute acid.

$$Zn + 2\ HCl \rightarrow ZnCl_2 + H_2\uparrow$$

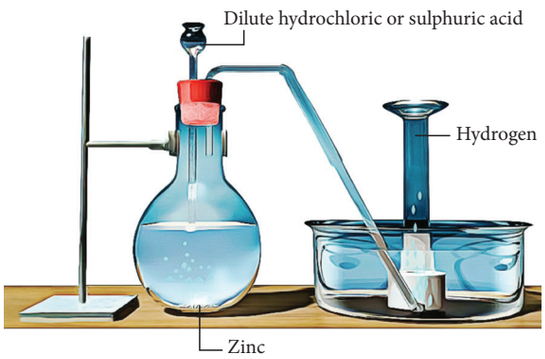



**Figure 4.2 Laboratory preparation of hydrogen**

### 4.2.2 Industrial Production

In the large-scale, hydrogen is produced by steam-reforming of hydrocarbons. In this method hydrocarbon such as methane is mixed with steam and passed over nickel catalyst in the range 800-900 °C and 35 atm pressures.

$$CH_4 + H_2O \xrightarrow{Ni} CO + 3H_2$$

In an another process, steam is passed over a red-hot coke to produce carbon monoxide and hydrogen. The mixture of gases produced in this way is known as *water gas* ($CO+H_2$). This is also called *syngas* (Synthetic gas) as it is used in the synthesis of organic compounds such as methanol and simple hydrocarbons.

$$C + H_2O \xrightarrow{1000^\circ C} \underbrace{CO + H_2}_{\text{Water gas/Syngas}}$$

### Conversion of Carbon monoxide in water gas to Carbon dioxide:

The carbon monoxide of the water gas can be converted to carbon dioxide by mixing the gas mixture with more steam at 400°C and passed over a shift converter containing iron/copper catalyst. This reaction is called as *water-gas shift reaction*.

$$CO + H_2O \rightarrow CO_2 + H_2$$

The $CO_2$ formed in the above process is absorbed in a solution of potassium carbonate.

$$CO_2 + K_2CO_3 + H_2O \rightarrow 2\ KHCO_3$$



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